02 feature engineering

7 de abril de 2026

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Summary

Notebook 2: Feature Engineering

Input: `outputs/rba_cleaned.parquet` (from Notebook 1)
Output: `outputs/rba_features.parquet`

Feature groups created:

- **Temporal** — hour, day-of-week, month, is_weekend, is_night
- **Geo** — country/region/city frequency encoding
- **Device/Browser/OS** — label encoding and top-N grouping
- **Network** — ASN frequency encoding
- **User behaviour** — user login frequency, user-country combinations
- **RTT** — normalized, log-transformed

In [30]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os, warnings, time
warnings.filterwarnings('ignore')

OUTPUT_DIR = os.path.join(os.getcwd(), 'outputs')
clean_path = os.path.join(OUTPUT_DIR, 'rba_cleaned.parquet')

if not os.path.exists(clean_path):
    raise FileNotFoundError(f'Missing: {clean_path}\nRun Notebook 1 first.')
print(f'File found: {os.path.getsize(clean_path)/1e6:.1f} MB')

t0 = time.time()
df = pd.read_parquet(clean_path, engine='pyarrow')
print(f'Loaded in {time.time()-t0:.1f}s  |  shape: {df.shape}')
df.head(3)
```

Out [30]:

```
File found: 47.1 MB
Loaded in 0.4s  |  shape: (1000000, 15)
```

	Login Timestamp	User ID	Round-Trip Time [ms]	IP Address	Country	Region	City	ASN	User Agent String	Brov
0	2020-02-03 12:46:25.828	-4324475583306591935	NaN	37.191.162.246	NO	Oslo County	Oslo	57963	Mozilla/5.0 (Android 6.0.99; Tablet;…	Chrome M
1	2020-02-03 12:46:52.905	-4324475583306591935	NaN	94.198.241.134	CZ	Hlavni mesto Praha	Prague	62160	Mozilla/5.0 (iPhone; CPU iPhone …	Chrome M

	Login Timestamp	User ID	Round-Trip Time [ms]	IP Address	Country	Region	City	ASN	User Agent String	Brov
2	2020-02-03 12:46:55.455	9187119392009558735	NaN	10.0.183.79	RO	Arges	Curtea de Argeş	500372	Mozilla/5.0 (X11; CrOS armv7l 121…	Chrome 6

1. Temporal Features

In [31]:

```
ts = df['Login Timestamp']

df['hour']       = ts.dt.hour
df['dayofweek']  = ts.dt.dayofweek        # 0=Monday, 6=Sunday
df['month']      = ts.dt.month
df['day']        = ts.dt.day
df['is_weekend'] = (df['dayofweek'] >= 5).astype(int)
df['is_night']   = ((df['hour'] >= 22) | (df['hour'] < 6)).astype(int)

print('Temporal features added.')
print(df[['hour', 'dayofweek', 'month', 'is_weekend', 'is_night']].describe())
```

Out [31]:

```
Temporal features added.
                hour       dayofweek           month      is_weekend  \
count  1000000.000000  1000000.000000  1000000.000000  1000000.000000
mean        12.220263        2.891040        6.377444        0.259222
std          5.679444        1.982303        3.723848        0.438208
min          0.000000        0.000000        1.000000        0.000000
25%          8.000000        1.000000        3.000000        0.000000
50%         12.000000        3.000000        6.000000        0.000000
75%         17.000000        5.000000       10.000000        1.000000
max         23.000000        6.000000       12.000000        1.000000

            is_night
count  1000000.000000
mean         0.172374
std          0.377705
min          0.000000
25%          0.000000
50%          0.000000
75%          0.000000
max          1.000000
```

In [ ]:

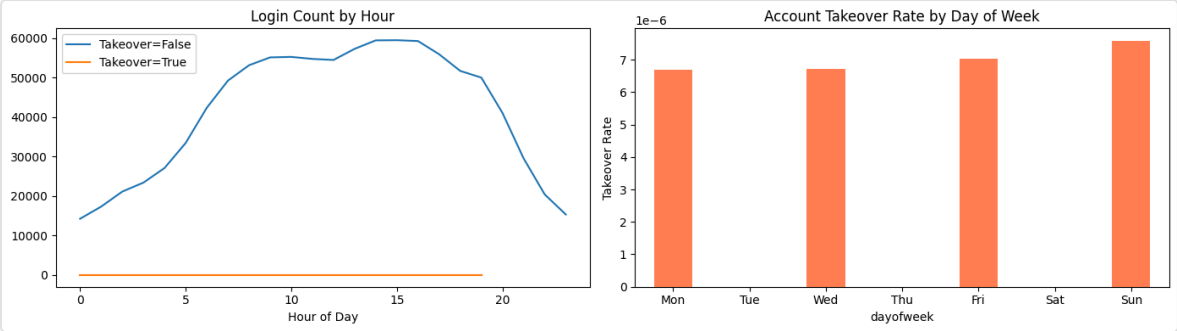
```
fig, axes = plt.subplots(1, 2, figsize=(14, 4))

# Logins by hour split by Is Attack IP
for label, grp in df.groupby('Is Attack IP'):
    grp['hour'].value_counts().sort_index().plot(ax=axes[0],
        label=f'Attack IP={label}', linewidth=1.5)
axes[0].set_title('Login Count by Hour (split by Is Attack IP)')
axes[0].set_xlabel('Hour of Day')
axes[0].legend()

# Attack IP rate by day-of-week
days = ['Mon','Tue','Wed','Thu','Fri','Sat','Sun']
df.groupby('dayofweek')['Is Attack IP'].mean().plot(
    kind='bar', ax=axes[1], color='tomato')
axes[1].set_title('Attack IP Rate by Day of Week')
axes[1].set_xticklabels(days, rotation=0)
axes[1].set_ylabel('Attack IP Rate')

plt.tight_layout()
plt.show()
```

Out [ ]:



2. Geo Features — Frequency Encoding

In [33]:

```
def freq_encode(df, col, new_col):
    freq_map = df[col].value_counts(normalize=True).to_dict()
    df[new_col] = df[col].map(freq_map).fillna(0)
    return df

df = freq_encode(df, 'Country', 'country_freq')
df = freq_encode(df, 'Region',  'region_freq')
df = freq_encode(df, 'City',    'city_freq')

print('Geo frequency features added.')
print(df[['country_freq', 'region_freq', 'city_freq']].describe())
```

Out [33]:

```
Geo frequency features added.
         country_freq     region_freq       city_freq
count  1000000.000000  1000000.000000  1000000.000000
mean         0.262593        0.218112        0.205618
std          0.167563        0.210006        0.215784
min          0.000001        0.000000        0.000000
25%          0.054077        0.015392        0.001800
50%          0.278001        0.073781        0.073513
75%          0.422239        0.449667        0.445555
max          0.422239        0.449667        0.445555
```

3. Network Feature — ASN Frequency Encoding

In [34]:

```
df = freq_encode(df, 'ASN', 'asn_freq')

# Top ASNs by attack rate
top_asn = (df.groupby('ASN')['Is Attack IP']
             .agg(['mean', 'count'])
             .query('count >= 100')
             .sort_values('mean', ascending=False)
             .head(15))
print('Top 15 ASNs by Attack IP rate (min 100 logins):')
print(top_asn)
```

Out [34]:

```
Top 15 ASNs by Attack IP rate (min 100 logins):
            mean  count
ASN
197535  1.000000    147
500203  1.000000    297
57423   0.973832    535
500254  0.942911   1594
500021  0.920698   1261
22612   0.873112    331
63673   0.816054    299
398986  0.781367   9907
41944   0.624079    407
```

```
501404  0.620915    153
265355  0.566667    150
500039  0.539764   5432
207674  0.532212   2794
134674  0.469256    309
500055  0.430000    100
```

4. Device / Browser / OS Features

In [35]:

```
# Device type: label encode
device_map = {v: i for i, v in enumerate(df['Device Type'].value_counts().index)}
df['device_type_enc'] = df['Device Type'].map(device_map).fillna(-1).astype(int)

# Browser: keep top 20, rest → 'Other'
top_browsers = df['Browser Name and Version'].value_counts().nlargest(20).index
df['browser_group'] = df['Browser Name and Version'].where(
    df['Browser Name and Version'].isin(top_browsers), 'Other')
df = freq_encode(df, 'browser_group', 'browser_freq')

# OS: keep top 20, rest → 'Other'
top_os = df['OS Name and Version'].value_counts().nlargest(20).index
df['os_group'] = df['OS Name and Version'].where(
    df['OS Name and Version'].isin(top_os), 'Other')
df = freq_encode(df, 'os_group', 'os_freq')

print('Device/Browser/OS features added.')
print('Device type encoding:')
print(dict(sorted(device_map.items(), key=lambda x: x[1])))
```

Out [35]:

```
Device/Browser/OS features added.
Device type encoding:
{'mobile': 0, 'desktop': 1, 'bot': 2, 'unknown': 3, 'tablet': 4}
```

In [ ]:

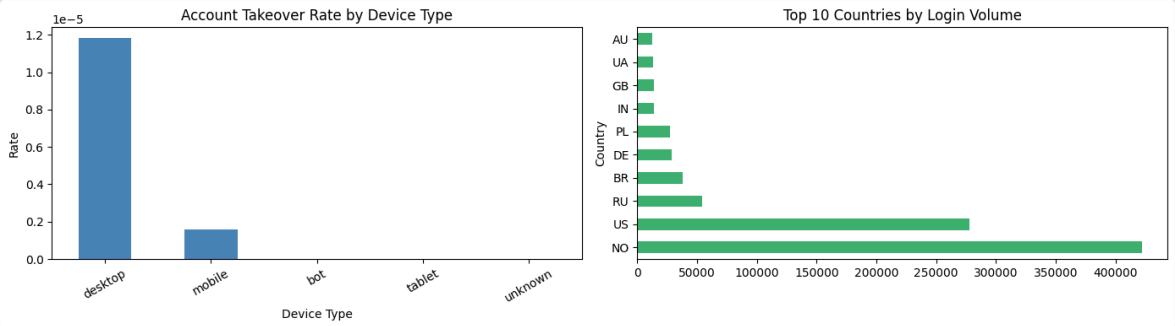
```
fig, axes = plt.subplots(1, 2, figsize=(14, 4))

# Attack IP rate by device type
(df.groupby('Device Type')['Is Attack IP'].mean()
   .sort_values(ascending=False)
   .plot(kind='bar', ax=axes[0], color='tomato'))
axes[0].set_title('Attack IP Rate by Device Type')
axes[0].set_ylabel('Rate')
axes[0].tick_params(axis='x', rotation=30)

# Top 10 countries by login volume
(df['Country'].value_counts().head(10)
   .plot(kind='barh', ax=axes[1], color='mediumseagreen'))
axes[1].set_title('Top 10 Countries by Login Volume')

plt.tight_layout()
plt.show()
```

Out [ ]:



5. User Behaviour Features

In [37]:

```
# User login frequency
user_freq = df['User ID'].value_counts().to_dict()
df['user_login_freq'] = df['User ID'].map(user_freq)

# Number of distinct countries per user
user_country_cnt = df.groupby('User ID')['Country'].nunique().to_dict()
df['user_country_count'] = df['User ID'].map(user_country_cnt)

# Number of distinct IPs per user
user_ip_cnt = df.groupby('User ID')['IP Address'].nunique().to_dict()
df['user_ip_count'] = df['User ID'].map(user_ip_cnt)

print('User behaviour features added.')
print(df[['user_login_freq', 'user_country_count', 'user_ip_count']].describe())
```

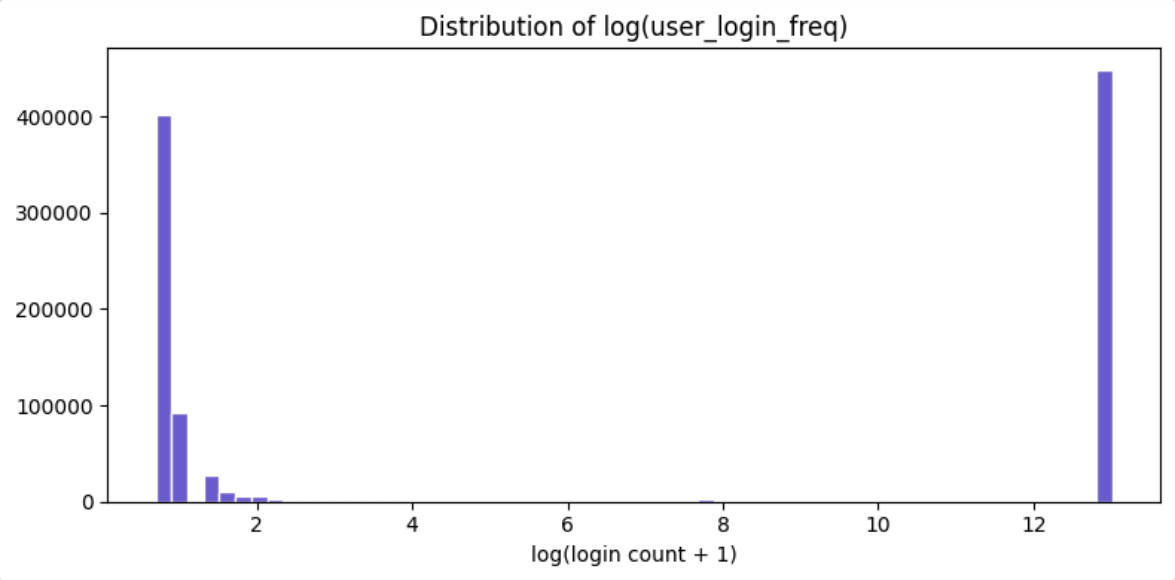
Out [37]:

```
User behaviour features added.
       user_login_freq  user_country_count   user_ip_count
count   1000000.000000      1000000.000000  1000000.000000
mean     201001.274792           92.014689    73316.232913
std      222956.876713          100.952312    81327.512691
min           1.000000            1.000000        1.000000
25%           1.000000            1.000000        1.000000
50%           3.000000            1.000000        2.000000
75%      448325.000000          204.000000   163532.000000
max      448325.000000          204.000000   163532.000000
```

In [38]:

```
fig, ax = plt.subplots(figsize=(8, 4))
ax.hist(np.log1p(df['user_login_freq']), bins=60, color='slateblue', edgecolor='white')
ax.set_title('Distribution of log(user_login_freq)')
ax.set_xlabel('log(login count + 1)')
plt.tight_layout()
plt.show()
```

Out [38]:



6. RTT Transformation

In [39]:

```
df['rtt_log1p'] = np.log1p(df['Round-Trip Time [ms]'].clip(lower=0))

# Z-score normalise
rtt_mean = df['rtt_log1p'].mean()
rtt_std  = df['rtt_log1p'].std()
df['rtt_zscore'] = (df['rtt_log1p'] - rtt_mean) / rtt_std

print(f'RTT log+zscore: mean={rtt_mean:.3f}, std={rtt_std:.3f}')
print(df[['Round-Trip Time [ms]', 'rtt_log1p', 'rtt_zscore']].describe())
```

Out [39]:

```
RTT log+zscore: mean=6.226, std=0.853
       Round-Trip Time [ms]     rtt_log1p    rtt_zscore
count          40563.000000  40563.000000  4.056300e+04
mean             672.323521      6.225518 -2.235171e-16
std             1189.353348      0.853245  1.000000e+00
min                8.000000      2.197225 -4.721144e+00
25%              471.000000      6.156979 -8.032790e-02
50%              538.000000      6.289716  7.523885e-02
75%              678.000000      6.520621  3.458592e-01
max            86587.000000     11.368917  6.028042e+00
```

6b. Security Behaviour Features

These features directly capture credential-stuffing and account-takeover patterns by summarising past activity per IP and per user. Computed from the same 1M-row sample — no external data needed.

In [40]:

```
# Sort by timestamp so lookbacks are temporally valid
df = df.sort_values('Login Timestamp').reset_index(drop=True)

# --- Per-IP aggregate features ---
ip_grp = df.groupby('IP Address')

# Total logins from the same IP
df['ip_login_count'] = df['IP Address'].map(ip_grp['Login Successful'].count())

# Failed logins from the same IP (aggregate)
ip_fail = ip_grp['Login Successful'].apply(lambda s: (~s.astype(bool)).sum()).to_dict()
df['ip_fail_count'] = df['IP Address'].map(ip_fail)

# Failure rate for the IP
df['ip_fail_rate'] = df['ip_fail_count'] / df['ip_login_count'].clip(lower=1)

# Distinct users seen from the same IP (many users → suspicious)
df['ip_user_count'] = df['IP Address'].map(ip_grp['User ID'].nunique())

# --- Cumulative previous counts per IP (temporally valid, no future leakage) ---
# Number of prior login attempts from this IP (0 = first time seen)
df['ip_prev_login_count'] = df.groupby('IP Address').cumcount()

# Binary fail flag used for both IP and user cumulative fail counts
df['_fail'] = (~df['Login Successful'].astype(bool)).astype(int)

# Number of prior failed attempts from this IP (excludes current row)
df['ip_prev_fail_count'] = df.groupby('IP Address')['_fail'].transform(
    lambda s: s.cumsum() - s
)

print('Per-IP security features added:')
print(df[['ip_login_count', 'ip_fail_count', 'ip_fail_rate',
          'ip_user_count', 'ip_prev_login_count', 'ip_prev_fail_count']].describe())
```

Out [40]:

```
Per-IP security features added:
       ip_login_count   ip_fail_count    ip_fail_rate   ip_user_count  \
count  1000000.000000  1000000.000000  1000000.000000  1000000.000000
mean       629.716200      620.000683        0.599152       14.990455
std       2256.132444     2257.904835        0.398660       42.500007
min          1.000000        0.000000        0.000000        1.000000
```

```
25%          1.000000        1.000000        0.142857        1.000000
50%          4.000000        2.000000        0.733333        1.000000
75%         90.000000       67.000000        1.000000       16.000000
max      14260.000000    14260.000000        1.000000      956.000000

       ip_prev_login_count  ip_prev_fail_count
count       1000000.000000      1000000.000000
mean            314.358100          309.527691
std            1315.202022         1315.820594
min               0.000000            0.000000
25%               0.000000            0.000000
50%               1.000000            0.000000
75%              45.000000           31.000000
max           14259.000000        14259.000000
```

In [41]:

```python
# --- Per-user aggregate features ---
user_grp = df.groupby('User ID')

# Historical login success rate per user (aggregate)
user_success = user_grp['Login Successful'].mean().to_dict()
df['user_prev_success_rate'] = df['User ID'].map(user_success)

# Time since previous login for the same user (seconds), -1 if first login
df['_prev_ts'] = df.groupby('User ID')['Login Timestamp'].shift(1)
df['user_time_since_last_login'] = (
    df['Login Timestamp'] - df['_prev_ts']
).dt.total_seconds().fillna(-1).clip(lower=-1)
df.drop(columns=['_prev_ts'], inplace=True)

# --- Cumulative previous counts per user (temporally valid) ---
# Number of prior logins by this user (0 = first login)
df['user_prev_login_count'] = df.groupby('User ID').cumcount()

# Number of prior failed logins by this user (reuses _fail from IP cell)
df['user_prev_fail_count'] = df.groupby('User ID')['_fail'].transform(
    lambda s: s.cumsum() - s
)
df.drop(columns=['_fail'], inplace=True)

# --- Context-change features (1 = changed from previous login by same user) ---
# Country changed since previous login
df['_prev_country'] = df.groupby('User ID')['Country'].shift(1)
df['country_changed'] = (
    (df['Country'] != df['_prev_country']) & df['_prev_country'].notna()
).astype(int)
df.drop(columns=['_prev_country'], inplace=True)

# Device type changed since previous login
df['_prev_device'] = df.groupby('User ID')['Device Type'].shift(1)
df['device_changed'] = (
    (df['Device Type'] != df['_prev_device']) & df['_prev_device'].notna()
).astype(int)
df.drop(columns=['_prev_device'], inplace=True)

print('Per-user security features added:')
print(df[['user_prev_success_rate', 'user_time_since_last_login',
          'user_prev_login_count', 'user_prev_fail_count',
          'country_changed', 'device_changed']].describe())
```

Out [41]:

```
Per-user security features added:
       user_prev_success_rate  user_time_since_last_login  \
count          1000000.000000                1.000000e+06
mean                 0.400848                5.372567e+05
std                  0.481050                2.703031e+06
min                  0.000000               -1.000000e+00
25%                  0.000000               -1.000000e+00
50%                  0.000000                4.874000e+00
75%                  1.000000                6.802900e+01
max                  1.000000                3.343596e+07

       user_prev_login_count  user_prev_fail_count  country_changed  \
```

```
count         1000000.000000        1000000.000000   1000000.000000
mean           100500.137396         100499.959498        0.338112
std            141197.446968         141197.573576        0.473067
min                 0.000000              0.000000        0.000000
25%                 0.000000              0.000000        0.000000
50%                 1.000000              0.000000        0.000000
75%            198324.250000         198324.250000        1.000000
max            448324.000000         448324.000000        1.000000

       device_changed
count  1000000.000000
mean         0.243170
std          0.428997
min          0.000000
25%          0.000000
50%          0.000000
75%          0.000000
max          1.000000
```

In [42]:

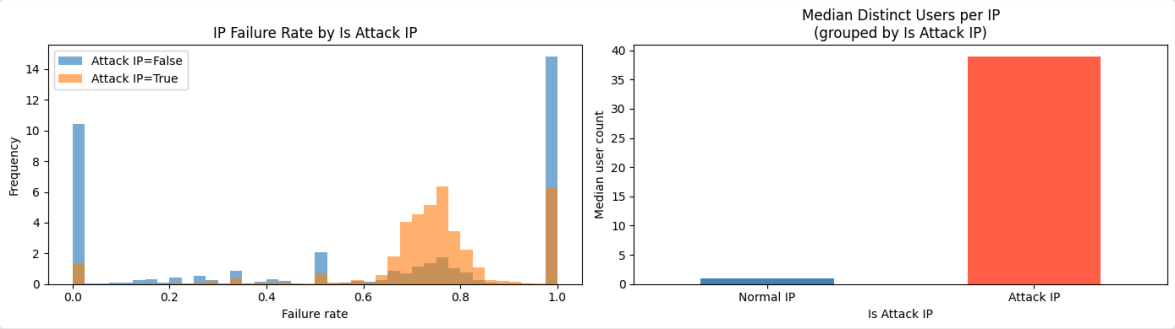
```
# Visualise: IP failure rate and user count split by Is Attack IP
fig, axes = plt.subplots(1, 2, figsize=(14, 4))

for label, grp in df.groupby('Is Attack IP'):
    grp['ip_fail_rate'].clip(0, 1).plot.hist(bins=40, alpha=0.6,
        label=f'Attack IP={label}', ax=axes[0], density=True)
axes[0].set_title('IP Failure Rate by Is Attack IP')
axes[0].set_xlabel('Failure rate')
axes[0].legend()

df.groupby('Is Attack IP')['ip_user_count'].median().plot(
    kind='bar', ax=axes[1], color=['steelblue', 'tomato'])
axes[1].set_title('Median Distinct Users per IP\n(grouped by Is Attack IP)')
axes[1].set_xticklabels(['Normal IP', 'Attack IP'], rotation=0)
axes[1].set_ylabel('Median user count')

plt.tight_layout()
plt.show()
```

Out [42]:



7. Select Final Feature Set

In [43]:

```
FEATURE_COLS = [
    # Temporal
    'hour', 'dayofweek', 'month', 'is_weekend', 'is_night',
    # Network / Geo
    'country_freq', 'region_freq', 'city_freq', 'asn_freq',
    # Device / Browser / OS
    'device_type_enc', 'browser_freq', 'os_freq',
    # RTT
    'rtt_log1p', 'rtt_zscore',
    # User behaviour (aggregate)
    'user_login_freq', 'user_country_count', 'user_ip_count',
    # Security behaviour — aggregate per IP
```

```
    'ip_login_count', 'ip_fail_count', 'ip_fail_rate', 'ip_user_count',
    # Security behaviour — user history (aggregate)
    'user_prev_success_rate',
    # Cumulative previous counts (temporally valid, no future leakage)
    'ip_prev_login_count', 'ip_prev_fail_count',
    'user_prev_login_count', 'user_prev_fail_count',
    # Time gap
    'user_time_since_last_login',
    # Context-change flags
    'country_changed', 'device_changed',
]

TARGET_COLS = ['Is Account Takeover', 'Is Attack IP', 'Login Successful']

df_features = df[FEATURE_COLS + TARGET_COLS].copy()
print(f'Feature matrix shape: {df_features.shape}')
print(f'Total features: {len(FEATURE_COLS)}')
df_features.head(3)
```

Out [43]:

```
Feature matrix shape: (1000000, 32)
Total features: 29
```

	hour	dayofweek	month	is_weekend	is_night	country_freq	region_freq	city_freq	asn_freq	device_type_enc	...	ip_prev_login_count	ip_prev_fail_count	user_prev_logir
0	12	0	2	0	0	0.422239	0.073781	0.073513	0.007960	0	...	0	0	0
1	12	0	2	0	0	0.003854	0.001255	0.001250	0.000628	0	...	0	0	1
2	12	0	2	0	0	0.003907	0.000030	0.000003	0.000020	1	...	0	0	0

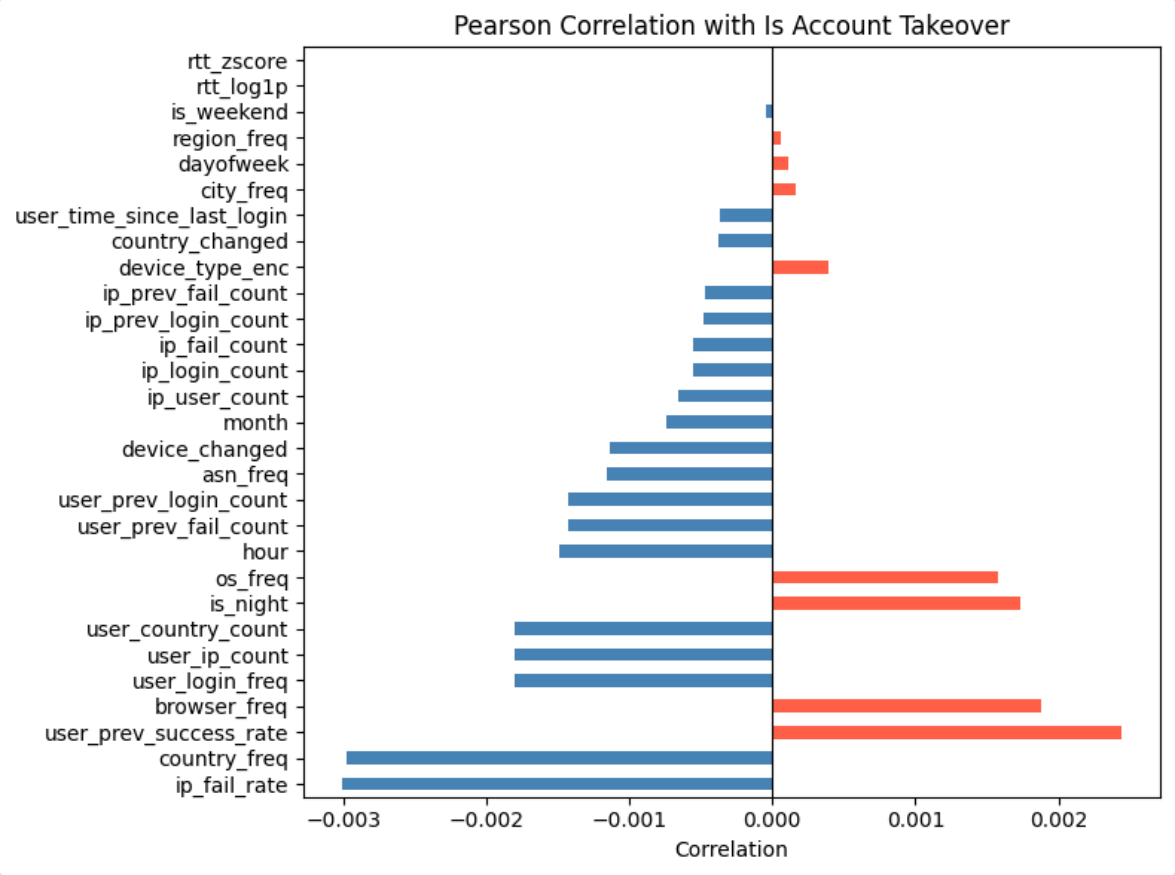
3 rows × 32 columns

In [ ]:

```
# Correlation heatmap (features vs primary target: Is Attack IP)
corr = df_features[FEATURE_COLS + ['Is Attack IP']].corr()
attack_corr = corr['Is Attack IP'].drop('Is Attack IP').sort_values(key=abs, ascending=False)

fig, ax = plt.subplots(figsize=(8, 9))
attack_corr.plot(kind='barh', ax=ax, color=[('tomato' if v > 0 else 'steelblue') for v in attack_corr])
ax.axvline(0, color='black', linewidth=0.8)
ax.set_title('Pearson Correlation with Is Attack IP (primary target)')
ax.set_xlabel('Correlation')
plt.tight_layout()
plt.show()
```

Out [ ]:



8. Save Feature Matrix

In [45]:

```
feat_path = os.path.join(OUTPUT_DIR, 'rba_features.parquet')
df_features.to_parquet(feat_path, index=False, engine='pyarrow')
print(f'Feature matrix saved: {feat_path}')
print(f'File size: {os.path.getsize(feat_path)/1e6:.1f} MB')

# Save the feature column list for use in later notebooks
import json
meta = {'feature_cols': FEATURE_COLS, 'target_cols': TARGET_COLS}
meta_path = os.path.join(OUTPUT_DIR, 'feature_meta.json')
with open(meta_path, 'w') as f:
    json.dump(meta, f, indent=2)
print(f'Feature metadata saved: {meta_path}')
```

Out [45]:

```
Feature matrix saved: c:\Users\ruchi\Downloads\rba-dataset.csv\outputs\rba_features.parquet
File size: 23.4 MB
Feature metadata saved: c:\Users\ruchi\Downloads\rba-dataset.csv\outputs\feature_meta.json
```

Summary

Feature Group	Features Created	Count
Temporal	hour, dayofweek, month, is_weekend, is_night	5
Geo / Network	country_freq, region_freq, city_freq, asn_freq	4
Device / Browser / OS	device_type_enc, browser_freq, os_freq	3
RTT	rtt_log1p, rtt_zscore	2
User behaviour (aggregate)	user_login_freq, user_country_count, user_ip_count	3
IP security (aggregate)	ip_login_count, ip_fail_count, ip_fail_rate, ip_user_count	4

Feature Group	Features Created	Count
User security (aggregate)	user_prev_success_rate	1
Cumulative prev. counts (temporally valid)	ip_prev_login_count, ip_prev_fail_count, user_prev_login_count, user_prev_fail_count	4
Time gap	user_time_since_last_login	1
Context-change flags	country_changed, device_changed	2

Total features: 29

→ Next: Notebook 3 — Baseline + Logistic Regression + Random Forest